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This is a valid value for a.

If $A = 11$, and $C = 8$, then

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 11a + 8(11 - a) &= 101 && \text{substitution} \\ 11a + 88 - 8a &= 101 && \text{distributive property} \\ 3a + 88 &= 101 && \text{combine like terms} \\ 3a &= 13 && \text{subtract 88 from both sides} \\ a &= 133 && \text{divide both sides by 3} \end{aligned}$$

Since a must be a nonnegative integer, 133 is not a valid value for a.

If $A = 11$ and $C = 10$, then

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 11a + 10(11 - a) &= 101 && \text{substitution} \\ 11a + 110 - 10a &= 101 && \text{distributive property} \\ a + 110 &= 101 && \text{combine like terms} \\ a &= -9 && \text{subtract 110 from both sides} \end{aligned}$$

Since a must be a nonnegative integer, −9 is not a valid value for a.

If $A = 12$ and $C = 11$, then

$$\begin{aligned} aA + (11 - a)C &= 101 && \text{given} \\ 12a + 11(11 - a) &= 101 && \text{substitution} \\ 12a + 121 - 11a &= 101 && \text{distributive property} \\ a + 121 &= 101 && \text{combine like terms} \\ a &= -20 && \text{subtract 121 from both sides} \end{aligned}$$

Since a must be a nonnegative integer, −20 is not a valid value for a.

***A*:**

From the work above, letting $A = 11$ and $C = 6$ gives the only valid value for a.

The correct answer is *11*.

***C*:**

From the work above, letting $A = 11$ and $C = 6$ gives the only valid value for a.

The correct answer is *6*.

Note that $A = 11$ and $C = 6$ are consistent with the given information. First, since $11 > 6$, $A > C$. Also, there exist

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Note that $A = 11$ and $C = 6$ are consistent with the given information. First, since 11 > 6, $A > C$. Also, there exist numbers of adult and child tickets sold yesterday, namely 25 and 8, respectively, such that 33 tickets (25 + 8 = 33) were sold yesterday and they generated revenue of (25)(11) + (8)(6) = 275 + 48 = 323 dollars. Furthermore, there exist numbers of adult and child tickets sold today, namely 32 and 12, respectively, such that 44 tickets (32 + 12 = 44) were sold yesterday and they generated revenue of (32)(11) + (12)(6) = 352 + 72 = 424 dollars.

•

Companies A and B are part of the same industry and are located in the same city. For Company A, the average (arithmetic mean) salary of its employees, in United Arab Emirates dirhams (AED), is 10,000 AED higher than that for Company B. However, more than half of the employees at Company A have salaries below the average for Company B.

Statement: If the average salary at Company B is ____1____, then the median salary at Company A is ____2____.

Select for *1* and for *2* the options that complete the statement so that it most accurately reflects the information provided. Make only two selections, one in each column.

1	2	
		greater than 100,000 AED
		less than 100,000 AED
		equal to 110,000 AED
		between 100,000 and 110,000 AED
		greater than 110,000 AED

Answer Explanation:

Infer

The statement that more than half of the employees at Company A have salaries below the average salary at Company B is equivalent to the median salary at Company A is less than the average salary at Company B.

Therefore, the information provided can be summarized as

i. $a = 10{,}000 + b$ and

ii. $m < b$,

where

a = average salary at Company A

b = average salary at Company B

m = median salary at Company A

***1*:**

The statement asserts that if b satisfies a certain numerical condition, then m satisfies a certain (possibly different) numerical condition. Because this statement involves only b and m, it is reasonable to begin by considering (ii), which asserts that b is the larger of the two numbers b and m. Since knowing that the larger of two numbers is less than some given value implies that the smaller of the two numbers is also less than that given value, it follows that if b is less than 100,000, then m is less than 100,000. Moreover, if one is so inclined, it is not difficult to check that no other choice of options for *1* and *2* is necessarily true. Note that (i) is essentially irrelevant in selecting the options that complete the statement.

The correct answer is *less than 100,000 AED*.

***2*:**

As shown above, the statement most accurately reflects the information provided for a certain choice of options in which the option for the second selection is the same as the option for the first selection.

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which the option for the second selection is the same as the option for the first selection.

The correct answer is *less than 100,000 AED.*

•

In comparing the results of a recent annual spring census to those of the previous year, biologists observed dramatic changes in the numbers of frogs of two species—Species X and Species Y—and an overall decrease in the combined number of frogs of these species. The biologists hypothesized that this decrease was caused by the unusually cold weather between the two censuses.

Select for A and for B the statements such that the biologists' hypothesis would have the most support if A is true and B is false. Make only two selections, one in each column.

A	B	
		Species X is susceptible to unusually cold weather, whereas Species Y is not.
		Species X has a later mating season than does Species Y.
		Both Species X and Species Y are susceptible to similar contaminants.
		The number of Species X frogs decreased from the previous spring census.
		The number of Species Y frogs decreased from the previous spring census.

Answer Explanation:

Evaluate

Because the stem asks us to consider the likelihood of something occurring when A is true but B is false, we must consider both selections together.

A and B:

The biologists hypothesized that the decrease in the size between two censuses of the combined population of Species X and Species Y was caused by unusually cold weather between the two censuses. What would support this hypothesis? One possibility would be evidence that both Species X and Species Y are susceptible to unusually cold weather. Notice, however, that this is not one of the options to choose from. The closest option to this is the one stating "Species X is susceptible to unusually cold weather, whereas Species Y is not." This would support the hypothesis if, but only if, it is not the case that Species Y decreased between the two censuses and did so for some reason other than unusually cold weather. If that were the case, it could be that this other reason is what caused the combined population to decline between the two censuses and not unusually cold weather. Thus, the hypothesis would be supported if it were true that Species X is susceptible to unusually cold weather but Species Y is not while it is false that the number of Species Y frogs decreased from the previous spring census (in other words, the number of Species Y frogs either did not change or increased).

The correct answer for A is *Species X is susceptible to unusually cold weather, whereas Species Y is not.*

The correct answer for B is *The number of Species Y frogs decreased from the previous spring census.*

•

A certain company defines its *annual labor turnover rate* as the number of employees who left the company during the year divided by the average of the number of employees on the first day of the year and the number of employees on the last day of that year. Last year, the company had a labor turnover rate of exactly 25%.

Assuming that the information above is true, select for *Statement 1* and for *Statement 2* the statements such that if Statement 1 is true, then Statement 2 must be true, but it could be the case that Statement 2 is true and Statement 1 is false. Make only two selections, one in each column.

Statement 1	Statement 2	
		Last year, more people left the company than began work at the company

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		Last year, more people left the company than began work at the company.
		Exactly 25% of the employees working at the company on the first day of last year left the company last year.
		More than 25% of the employees working at the company on the first day of last year left the company last year.
		All and only those who were employees at the company on the first day of last year were employees on the last day of last year.
		Last year, the number of employees on the first day of the year was equal to the number of employees on the last day of the year.

Answer Explanation:

Evaluate

Note that because Statement 1 and Statement 2 must be chosen by determining whether a certain relationship between them holds—namely, that if Statement 1 is true, then Statement 2 must be true, but if Statement 2 is true, Statement 1 can be false—the two answer selections must be chosen together.

***Statement 1* and *Statement 2*:**

We are told that a certain company "defines its *annual labor turnover rate* as the number of employees who left the company during the year divided by the average of the number of employees on the first day of the year and the number of employees on the last day of that year." We are also told that the labor turnover rate last year was 25%. To answer this question correctly, you must find two statements such that if one of them (Statement 1) is true, then the other (Statement 2) must be true, but the reverse may not hold (i.e., if Statement 2 is true, Statement 1 can be false). Consider the fourth answer choice, "All and only those who were employees at the company on the first day of last year were employees on the last day of last year." If that is true, that means that last year, the number of employees on the first day of the year was equal to the number of employees on the last day of the year. Notice that that is what the fifth answer choice says. So, the fourth and fifth answer choices are likely candidates for the answer. (Note that it may seem odd that the company would have exactly the same employees on the first and last day of the year, given that the company had a 25% annual turnover rate. But that turnover rate might occur if, for instance, the company hired some workers during the year for a short-term project, and all those workers who were hired at some point after the first day were no longer employed there on the last day.) So, we've seen that if the fourth answer choice is true, the fifth answer choice must also be true. But does the fourth have to be true if the fifth is? No, because some people who were employees on the first of the year could have left during the year, and an equal number of employees were hired to replace them, all of whom remained employees on the last day of the year. That would result in the first and last days of the year having an equal number of employees, though not all the employees on the first day were employees on the last day.

The correct answer for Statement 1 is *All and only those who were employees at the company on the first day of last year were employees on the last day of last year.*

The correct answer for Statement 2 *is Last year, the number of employees on the first day was equal to the number of employees on the last day of the year.*

•

Newspaper editor: Published photojournalism must always present the events covered without distorting those events. Sometimes an image distorts the events a photojournalist attempts to capture, such as when the photojournalist has an equipment failure. In such cases, photojournalists are permitted to make minimal changes to an image, but only to the extent that (1) they are certain about what they observed, and (2) those changes reduce the distortion. Photojournalists' photos are often compelling works of art as well as documents of newsworthy events. However, for publication as journalism, a photo's aesthetic features are allowed to be considered only after the photo is shown to accurately portray relevant features of the event it depicts. In accordance with these criteria, in our journalistic coverage of Event E, we elected to publish photograph P1 rather than photograph P2.

Statement: Assuming the editor's statements are true, if ____1____, then it must be the case that ____2____.

Select for *1* and for *2* the two different options that create the statement that is most strongly supported by the

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Select for *1* and for *2* the two different options that create the statement that is most strongly supported by the information provided. Make only two selections, one in each column.

1	2	
		P1 was altered
		P2 was altered
		P1 and P2 were taken with malfunctioning equipment
		P1 was chosen over P2 at least in part for its aesthetic features
		both P1 and P2 accurately portray the relevant features of Event E

Answer Explanation:

Apply

Note that because the two responses are related to each other (they are both parts of a single conditional statement), the correct selections must be chosen together.

***1* and *2*:**

The newspaper editor's statements discuss a condition that must be met for photojournalism to be published—namely, the photos must accurately present the events without distorting the events—and when photojournalists are allowed to make changes to a photo—namely, when the photojournalists are certain about what they observed and the changes reduce the distortion. Some of the possible answer choices for this question might seem promising in light of these conditions expressed by the editor, but in each case, there isn't another appropriate answer choice to pair with it when based on those statements from the editor. For instance, the first answer choice, "P1 was altered," could be the answer for *1* if another answer choice was "The alteration of P1 reduced the distortion," but that is not an available answer choice. Instead, the correct answer choices to this question are based on another condition expressed by the editor, namely that a photo's aesthetic features can be considered only after determining that the photo accurately portrays the relevant features of the event it captures. This suggests that the fourth and fifth answer choices are good candidates as answers for *1* and *2*. If P1 was chosen for publication over P2, and that choice was based at least in part on P1's aesthetic features, then it must be that both P1 and P2 accurately portray the relevant features of Event E because to be published at all, P1 must accurately portray those features, and if P2 didn't accurately portray them as well, then the aesthetic features of P1 wouldn't have been relevant in choosing between P1 and P2 because P2 simply could not be published.

The correct answer for 1 is *P1 was chosen over P2 at least in part for its aesthetic features*.

The correct answer for 2 is *both P1 and P2 accurately portray the relevant features of Event E*.

•

A stock trader working for a hedge fund has estimated that the stock of a certain company has probability 0.4 of increasing in price by at least 5 dollars during a certain trading day and probability 0.1 of increasing in price by at least 10 dollars during the trading day.

Based on the trader's estimates, select for X and for Y the options such that the following statement is most accurate. Make only two selections, one in each column.

If the trader multiplies ____X____ by the reciprocal of ____Y____, then the result is the probability that the price of the stock will increase by at least 10 dollars during the trading day, given that the price increases by at least 5 dollars during the trading day.

X	Y	
		0.1
		0.4

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		0.1 divided by 0.4
		the reciprocal of 0.1
		the product of 0.1 and 0.4

Answer Explanation:

Strategize

Let Event A be "the stock increases in price by at least 5 dollars on a certain day," and let Event B be "the stock increases in price by at least 10 dollars on that day." The probability of B given A is $\frac{(\text{the probability of A and B.})}{(\text{the probability of A})}$.

The probability of A is given as 0.4. The probability of A and B is the same as the probability of B because Event A is contained in Event B (i.e., if the stock increased in price by at least $10, then the price certainly increased by at least $5). Therefore, the probability of A and B is the same as the probability of B, which is 0.1.

X:

The correct answer is *0.1*.

Y:

Because dividing by a number is equivalent to multiplying by the reciprocal of that number, the probability of A and B divided by the probability of A is equivalent to the probability of A and B multiplied by the reciprocal of the probability of A.

The correct answer is *0.4*.

•

Throughout a certain decade in a European city, the mean monthly rents for studio apartments varied yearly, from a low of €804 to a high of €1,173. Those for one-bedroom apartments also varied yearly, from a low of €1,060 to a high of €1,497. But some individual studio apartments rented for as little as €420 in some years, and some one-bedroom apartments rented for up to €2,262. To visually assess how the ratio of mean monthly rents for studio apartments to those for one-bedroom apartments varied yearly over the decade, Maria requires a graph with the following characteristics. The graph will have two axes of equal length, with mean rents for one-bedroom apartments shown on the horizontal axis and mean rents for studio apartments shown on the vertical axis, and with the same scale on both axes. For each year, mean rents will be plotted as a point.

From the following options, select for *Horizontal axis* a range for the points on the horizontal axis and select for *Vertical axis* a range for the points on the vertical axis that together would satisfy Maria's requirements for the graph. Make only two selections, one in each column.

Horizontal axis	Vertical axis	
		€0 to €1,500
		€400 to €1,100
		€800 to €1,200
		€1,000 to €2,300
		€1,100 to €1,500

Answer Explanation:

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Strategize

Because both axes must be the same length and have the same scale, the range for both axes must start low enough to accommodate €804 and go high enough to accommodate €1,497. The range €0 to €1,500 accommodates both the low of €804 and the high of €1,497. Neither of the ranges €400 to €1,100 and €800 to €1,200 goes high enough to accommodate €1,497, and neither of the ranges €1,000 to €2,300 and €1,100 to €1,500 goes low enough to accommodate €804.

***Horizontal axis*:**

The correct answer is *€0 to €1,500*.

***Vertical axis*:**

The correct answer is *€0 to €1,500*.

•

A statistician reached the following conclusions about games between university soccer teams: Overall, a team playing on its home field has a 45% chance of a win, a 25% chance of a loss, and a 30% chance of a draw (a tied outcome). In the games where one or more goals are scored, the team that scores the first goal has a 55% chance of scoring it in the game's first half and a 45% chance of scoring it later in the game. When that team is the home team (i.e., a team playing on its home field), there is a 40% chance that the other team will score no goals at all, and therefore a 60% chance that it will score one or more goals.

Select for X and for Y two different outcomes such that the information provided explicitly includes the statistician's estimates of the probability that if X occurs, so will Y. Make only two selections, one in each column.

X	Y	
		The home team scores at least two goals in the game.
		The home team scores the first goal.
		A goal is scored in the first half of the game.
		A goal is scored in the second half of the game.
		The team opposing the home team scores at least one goal.

Answer Explanation:

Recognize

The key information for determining outcomes X and Y such that if X occurs, then so does Y is "When that team (where 'that team' refers to the team that scores the first goal) is the home team …, there is … a 60% chance that it (where 'it' refers to the other team) will score one or more goals." Therefore, from the given information, if the home team scores the first goal, the statistician's estimate of the probability that the team opposing the home team scores at least one goal is 0.6.

X:

The correct answer is *The home team scores the first goal*.

Y:

The correct answer is *The team opposing the home team scores at least one goal*.

•

To ensure computer security, a firm has rules about access of managerial and nonmanagerial employees to various networked computer drives. In accordance with the rules, some but not all nonmanagerial employees are allowed access to both Drive B and Drive D. Some are required to have such access. Every nonmanagerial employee must have access to at least one of Drive B and Drive D. No nonmanagerial employee is allowed to have access to both Drive B and Drive E. The rules do not restrict the access of managerial employees to any of the drives mentioned.

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From the following statements about computer security practices in relation to the firm's employees, select for *Required* the statement that, based on the information provided, describes a practice that is required by the rules, and select for *Permitted* the statement that describes a practice that is permitted, but NOT required, by the rules. Make only two selections, one in each column.

Required	Permitted	
		At least one nonmanagerial employee is currently not allowed access to any of the three drives B, D, and E.
		Managerial employees, and only those, are currently allowed access to Drive E.
		The nonmanagerial employees who are currently allowed access to Drive D are also allowed access to Drive E.
		The managerial and nonmanagerial employees who are currently allowed access to Drive B are also allowed access to Drive E.
		Any nonmanagerial employees currently allowed access to Drive E are also allowed access to Drive D.

Answer Explanation:

Infer

***Required*:**

Since each nonmanagerial employee is allowed access to Drive B or Drive D, regardless of whether the employee is allowed access to Drive E, it follows that each nonmanagerial employee who is allowed access to Drive E is allowed access to Drive B or Drive D. Because no nonmanagerial employee is allowed access to both Drive B and Drive E, it follows that each nonmanagerial employee who is allowed access to Drive E is also allowed access to Drive D.

The correct answer is *Any nonmanagerial employees currently allowed access to Drive E are also allowed access to Drive D*.

***Permitted*:**

The rules do not restrict access to Drive E by managerial employees, so managerial employees are allowed access to Drive E. While the rules include a provision regarding access to Drive E by nonmanagerial employees, none of the rules requires that there exists a nonmanagerial employee who is allowed access to Drive E.

The correct answer is *Managerial employees, and only those, are currently allowed access to Drive E*.

•

Giulia is planning to sell her car, which is fueled by gasoline (petrol) and averages 20 miles per gallon (mpg), and purchase a diesel-fueled car that averages 30 mpg. She estimates that her future cost per gallon of diesel fuel will be 5% higher than her present cost per gallon of gasoline. She wishes to estimate (1) the annual cost of fuel for her new car if she maintains her present annual total miles driven and (2) the annual total miles she can drive her new car if she maintains her present annual expenditure on fuel.

Let x represent Giulia's present annual cost per gallon of gasoline in US dollars, and let y equal her present annual total of miles driven. Select for *Cost* an appropriate expression for Giulia's estimate of (1) above, and select for *Miles* an appropriate expression for her estimate of (2) above. Make only two selections, one in each column.

Cost	Miles	
		$\frac{2}{3}(1.05x)$
		$3(\ \ y\ \ )$

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		$\frac{3}{2}\left(\frac{y}{1.05}\right)$
		$\frac{3}{2}(1.05y)$
		$\frac{1.05xy}{30}$
		$\frac{1.05xy}{20}$
		$\frac{20xy}{1.05}$

Answer Explanation:

Strategize

***Cost*:**

The annual cost of fuel for her new car is (price per gallon)(number of gallons per year).

For her new car, the price of fuel per gallon will be 5% greater than the price of fuel per gallon for her present car. Given that x represents the price of fuel per gallon for her present car, the price of fuel per gallon for her new car is represented by $1.05x$.

If her new car averages 30 miles per gallon of fuel and she drives y miles annually, she will use $\frac{y}{30}$ gallons of fuel per year. Thus, the annual cost of fuel for her new car is represented by $1.05x\left(\frac{y}{30}\right)$.

The correct answer is $\frac{1.05\,xy}{30}$.

***Miles*:**

Her present annual expenditure on fuel is $x\frac{y}{20} = \frac{xy}{20}$. If T represents the total miles she can drive her new car while maintaining her present annual expenditure on fuel, then $(1.05x)\left(\frac{T}{30}\right) = \frac{xy}{20}$, from which it follows that

$T = \left(\frac{30}{1.05x}\right)\left(\frac{xy}{20}\right) = \frac{3}{2}\left(\frac{y}{1.05}\right)$.

The correct answer is $\frac{3}{2}\left(\frac{y}{1.05}\right)$.

•

Each of three botanists made a hypothesis regarding specimens of a particular plant species:

Botanist 1: Any individual specimen possessing the gene for curly stems has either the gene for long roots or the gene for purple flowers, or both.

Botanist 2: Any individual specimen possessing the gene for long roots has either the gene for flat leaves or the gene for round seeds, or both.

Botanist 3: No individual specimen that possesses either the gene for curly stems or the gene for flat leaves or both has the gene for purple flowers.

The discovery of an individual specimen of the plant species in question having the gene for ____1____ but NOT the gene for ____2____ would show that at least one of the three hypotheses described is incorrect.

Select for *1* and for *2* the characteristics that would most accurately complete the statement, based on the information

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Select for *1* and for *2* the characteristics that would most accurately complete the statement, based on the information given. Make only two selections, one in each column.

1	2	
		curly stems
		flat leaves
		long roots
		purple flowers
		round seeds

Answer Explanation:

Apply

***1*:**

If an individual specimen had the gene for curly stems and not the gene for long roots, then the hypothesis of Botanist 1 implies that the specimen would have the gene for purple flowers, and hence, by the hypothesis of Botanist 3, the specimen would not have the gene for curly stems (and also would not have the gene for flat leaves), which contradicts the fact that the specimen has the gene for curly stems. Therefore, such a specimen would show that at least one of the three hypotheses is incorrect. More precisely, such a specimen would show that at least one of the hypotheses of Botanist 1 and Botanist 3 is incorrect.

The correct answer is *curly stems*.

***2*:**

As shown in the explanation above, the specimen having the gene for curly stems and not having the gene for long roots would show that at least one of the three hypotheses is incorrect.

The correct answer is *long roots*.

•

An inventory of a neighborhood's trees found that 32 percent were conifers and most of the rest were deciduous. Among the conifers were 258 spruces and 112 pines, along with some cedars and other species. Most of the deciduous trees were oaks, but one in eight was a maple. Of the oaks, 65 percent were red oaks and 25 percent were white oaks. Of the maples, 20 percent were Japanese maples.

Select for A and for B two types of trees such that the ratio of the number of trees of the type selected for A to the number of trees of the type selected for B can be determined and is less than 1. Make only two selections, one in each column.

A	B	
		cedars
		conifers
		deciduous trees
		Japanese maples
		red oaks
		spruces

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		spruces

Answer Explanation:

Apply

Let T, C, D, O, M, and J be, respectively, the total number of trees in the neighborhood, the number of conifers, the number of deciduous trees, the number of oak trees, the number of maple trees, and the number of Japanese maple trees. It is given that $C = 0.32T$, $D > 0.50(1 - 0.32T)$ since "most of the rest were deciduous," and $O > 0.50D$ since "most of the deciduous trees were oaks." Also, it is given that $M = \frac{1}{8}D$ and $J = 0.20M$. It follows that $J = 0.20\left(\frac{1}{8}D\right) = \frac{1}{40}D$. Therefore, $JD = \frac{1}{40}$ and $\frac{1}{40} < 1$.

***A*:**

From the explanation above, the ratio $\frac{\text{Japanese}}{\text{deciduous}}$ can be determined and is less than 1.

The correct answer is *Japanese*.

***B*:**

From the explanation above, the ratio $\frac{\text{Japanese}}{\text{deciduous}}$ can be determined and is less than 1.

The correct answer is *deciduous*.

Alternatively, using the answer choices, cedars can be eliminated since "some cedars" doesn't give any precise amount to determine a ratio with any of the other answer choices. Although conifers make up 32 percent of all the trees in the neighborhood and there are 258 spruces, this information is not specific enough to determine a ratio with conifers and spruces. Since there is no information to determine a ratio between spruces and any of the other answer choices, spruces can be eliminated from further consideration. There is also no information to pair deciduous trees, which include Japanese maples and red oaks, with conifers. Thus, conifers can be eliminated from further consideration. Of the three answer choices remaining—deciduous trees, Japanese maples, and red oaks—there is insufficient information to determine a ratio between red oaks and deciduous trees because, even though red oaks comprise 65 percent of all oaks, there is no information as to how many oaks or how many deciduous trees are in the neighborhood. This leaves Japanese maples, which comprise 20 percent of the maples, and deciduous trees, one out of eight of which is a maple. Thus, Japanese maples = 0.2(maples) = 0.2 $\left(\frac{1}{8}\text{ deciduous}\right) = \frac{1}{40}$ deciduous. It follows that $\frac{\text{Japanese maples}}{\text{deciduous trees}} = \frac{1}{40}$ and $\frac{1}{40} < 1$.

•

A manufacturing company plans to begin automating production and reducing its workforce by installing industrial robots at the start of each year over five years. Each robot will result in annual labor-cost savings of €150,000 beginning with the year of its installation and will have an up-front cost of €400,000. For robot installation in Year 1 of the program, the company will budget €600,000, plus the total amount of labor costs that will be saved by robotic production in that year. The robot-installation budget for each subsequent year will consist of the total amount of labor costs that will be saved by robotic production in that year plus any money left over from the previous year's robot-installation budget.

Statement: Within budget constraints, the maximum number of robots that can be installed in Year 1 of the program is ____X____, and the maximum number of robots that can be in service in Year 3 is ____Y____.

Select values for X and for Y that create the statement that follows logically from the information provided. Make only two selections, one in each column.

X	Y	
		2

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		3
		4
		5
		6
		7

Answer Explanation:

Apply

Let x be the number of robots that can be installed in Year 1. Then

$$\begin{aligned} 600{,}000 + 150{,}000x &\geq 400{,}000x \\ 600{,}000 &\geq 250{,}000x \\ \frac{600{,}000}{250{,}000} &\geq x \\ 2.4 &\geq x \end{aligned}$$

The maximum number of robots that can be installed in Year 1 is 2. The budget remaining to be carried over to Year 2 is 600,000 + 2(150,000) − 2(400,000) = 100,000.

Assuming that at the beginning of Year 2 there are 2 robots working, let t be the number of robots that can be installed in Year 2. Then

$$\begin{aligned} 100{,}000 + 2(150{,}000) + 150{,}000t &\geq 400{,}000t \\ 400{,}000 &\geq 250{,}000t \\ \frac{400{,}000}{250{,}000} &\geq t \\ 1.6 &\geq t \end{aligned}$$

The maximum number of robots that can be installed in Year 2 is 1. The budget remaining to be carried over to Year 3 is 100,000 + 3(150,000) − 400,000 = 150,000.

Assuming that at the beginning of Year 3 there are 2 + 1 = 3 robots working, let y be the number of robots that can be installed in Year 3. Then

$$\begin{aligned} 150{,}000 + 3(150{,}000) + 150{,}000y &\geq 400{,}00y \\ 600{,}000 &\geq 250{,}000y \\ \frac{600{,}000}{250{,}000} &\geq y \\ 2.4 &\geq y \end{aligned}$$

The maximum number of robots that can be installed in Year 3 is 2.

***X*:**

From the explanation above, the maximum number of robots that can be installed in Year 1 is 2.

The correct answer is *2*.

***Y*:**

From the explanation above, the maximum numbers of robots that can be installed in Years 1, 2, and 3, respectively, are 2, 1, and 2. The maximum number of robots that can be in service in Year 3 is 2 + 1 + 2 = 5.

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are 2, 1, and 2. The maximum number of robots that can be in service in Year 3 is 2 + 1 + 2 = 5.

The correct answer is 5.

•

While designing a game involving chance, Desmond noticed that the probability a fair coin lands face up exactly 2 times when the coin is tossed 3 times is equal to the probability that a fair coin lands face up exactly m times when the coin is tossed n times, where $n > 3$.

Select for m and for n values consistent with the given information. Make only two selections, one in each column.

m	n	
		1
		2
		3
		4
		5
		6

Answer Explanation:

Infer

m:

If a coin is tossed 3 times and each toss can land only one of face up (U) or face down (D), then there are 3 possible outcomes where the coin lands face up exactly 2 times: UUD, UDU, and DUU. Here the notation is such that UDU, for example, means the coin landed face up, face

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